Michelle Dominic

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Technical Skills

Languages: Python, C, C++, C#, JavaScript, TypeScript, Java, SQL, SystemVerilog, VHDL, HTML/CSS
Frameworks & Libraries: FastAPI, Flask, Node.js, React, React Native, Vue.js, .NET, TensorFlow, Hugging Face
Embedded & Hardware: QNX, ESP32, STM32, Raspberry Pi, FPGA Design, PWM, UART, KiCad
Tools & Platforms: Git, Docker, Postman, MongoDB, Supabase, REST APIs, Azure Application Insights, Power BI, Quartus Prime, ModelSim
Certifications: React JS (Scalar Topics), SQL (HackerRank)

Education

University of Waterloo — Waterloo, ON, Canada
Candidate for BASc in Computer Engineering — *2024–2029*

Experience

Product Management Intern — May 2026 – Aug. 2026
SAP — *Waterloo, ON*

- Prioritized Administration & Monitoring features for **SAP HANA Cloud** by synthesizing customer requests, engineering constraints, and product strategy to define MVP scope
- Applied **Azure OpenAI embeddings** to cluster customer prompts and identify opportunities for specialized AI agents, informing multi-agent product planning
- Performed competitive analysis of AI model capabilities and pricing to evaluate trade-offs and support product roadmap decisions
- Partnered with customers, engineering, and product leadership to validate MVP requirements through customer engagement initiatives, preview testing, and feedback sessions
- Analyzed product adoption using **Azure Application Insights**, delivered internal workshops on customer pain points, and supported partner launch readiness through technical validation and documentation

FPGA Design Member — Jul. 2026 – Present
UWASIC — *Waterloo, ON*

- Developing **SystemVerilog RTL** for a **10 Gbps Ethernet packet parser** targeting FPGA hardware as part of the **IP packet parser** team
- Designing the **IP header extraction** stage, extracting IPv4 header fields from streaming packet data into structured metadata registers for downstream packet-processing modules
- Designing **verification logic** to validate packet parsing, field extraction, and metadata generation for IPv4 packets
- Collaborating with cross-functional parser teams using Git-based workflows, code reviews, and simulation-driven development

Product Management Intern — Sept 2025 – Dec 2025
SAP — *Waterloo, ON*

- Created technical tutorials for **HANA Cloud Central**, **HANA Client**, & **Knowledge Graph Engine**, including a high-traffic tutorial with **11,000+ visitors** supporting developer onboarding
- Supported **SAP HANA Cloud** product development through roadmap planning, & stakeholder coordination
- Analyzed **HANA Client** download data using **Python (Pandas)** & **Excel** to understand client usage patterns, & conducted a comparative analysis of competing client offerings
- Built a **usage analytics dashboard** with **Azure Application Insights** to track engagement across **HANA Cloud Central**

Full Stack Developer Intern — Jan. 2025 – Apr 2025
Stubbe's Precast — *Harley, ON*

- Created pipelines to train & test models using **Python**, **Excel**, & **SQL**, cutting data preparation time by **35%**
- Built & deployed **predictive models** using **TensorFlow**, **XGBoost**, & **Scikit-learn** to forecast warehouse production timelines, reducing error by **15%**
- Integrated an AI chatbot into internal **Digital Manufacturing** software using **C#**, **.NET**, **Node.js**, & **Vue.js**

Research Assistant May 2023 – Mar. 2024
University of Western Ontario *London, ON*

- Developed a **3D-printing G-code correction algorithm** in **FullControl**, improving structural integrity by **20%** & reducing material waste
- Generated economic heatmaps using **Python** & **Matplotlib** to identify cost-effective renewable strategies
- Contributed to **open-source** projects focused on energy optimization & sustainability

Projects

hand.shake: AI Grip-Assist Glove | *QNX, Raspberry Pi 5, ESP32, Python, Gemini API, Computer Vision* Jul. 2026

- Designed and assembled an **adjustable AI-powered wearable assistive glove** by integrating a **Raspberry Pi 5**, Camera Module 3 Wide, **ESP32**, tendon-driven actuation, and custom-mounted electronics into a compact prototype
- Engineered a **QNX 8** vision pipeline integrating **Gemini** with automatic fallback vision-language models, providing reliable object classification and grip-force estimation through resilient inference
- Developed **ESP32 firmware** to control tendon-actuated **360° servo motors via PWM**, receive wireless grip-force targets, and stream real-time telemetry to the Raspberry Pi
- Built a live telemetry dashboard visualizing detected objects, target versus applied grip force, and force-response curves for real-time system validation and debugging

MeshGit | *React, Three.js, Node.js, Express.js, Python, PostgreSQL, Docker* 2026

- Built a **geometry-aware version control system** for 3D STL mesh files, rendering visual diffs (added/removed/unchanged geometry) in an interactive **Three.js** viewport instead of text-based diffs
- Designed a microservice architecture separating a **Node.js/Express** API from a **Python** geometry service for mesh cleaning, boolean operations, and diffing on upload
- Implemented commit history, branching, **GitHub repo integration**, and conflict-zone identification for overlapping geometry merges, backed by **PostgreSQL** and containerized with **Docker Compose**

ElevateHER (Hackathon) | *React Native, Node.js, Express.js, MongoDB, Gemini AI* Sept. 2025

- Built a React Native mobile app for **peer mentorship & skill-sharing** using a swipe-based matching interface
- Integrated **AI-powered recommendations** using the **Gemini API** to match mentors & mentees
- Developed backend services using **Node.js**, **Express.js**, & **MongoDB** to manage users & mentorship sessions

Summus | *JavaScript, HTML/CSS, FastAPI, Python, Postman* Apr. 2025

- Built a **Chrome extension** summarizing terms & conditions, using **JavaScript** & **HTML/CSS** for frontend
- Engineered a backend with **FastAPI** & **Flask-CORS**, validating **REST** endpoints via **Postman**
- Ran local **LLM** inference via **Ollama** & **Hugging Face Transformers**, embedding a contextual **chatbot**

ASL Recognition System | *Python, TensorFlow/Keras, FastAPI, Node.js, React, Docker* Jan. 2025

- Training a **CNN image classifier** to recognize static ASL hand signs from a labeled dataset
- Building a **FastAPI ML inference service** to preprocess images & return predicted letters
- Implementing a **Node.js API gateway** to handle image uploads & forward requests to the ML service
- Containerizing services with **Docker Compose** to support a modular, multi-service architecture
- Planning **hardware-agnostic input support**, including future integration with **ESP32-CAM** devices

Credit+ | *Python, React, FastAPI, PostgreSQL, Three.js, WebXR, LangChain* Oct. 2025

- Designing **gamified financial learning** with lessons, quizzes, streaks & badges for personal finance education
- Implemented **backend tracking** (FastAPI + PostgreSQL) for user streaks & personalized learning paths
- Building an **AI financial** coach using **LangChain** & a **RAG** pipeline for tailored feedback on spending habits

Home Energy Monitor & HVAC Controller | *VHDL, Quartus Prime, ModelSim, Intel MAX FPGA* May 2025

- Designed & implemented an FPGA-based temperature control system on an **Intel MAX 10**, comparing real-time temperature inputs against user-defined set-points to drive HVAC control logic
- Developed modular **VHDL** components with synchronous control logic, verified functionality through unit-level & top-level simulations in **ModelSim**, & integrated the design using **Quartus Prime**
- Implemented real-time system feedback via dual seven-segment displays, mapping internal state & temperature values to hardware outputs

SenseSecure: Adaptive Alarm System | *C, STM32, KiCad, UART* Dec. 2024

- Developed an adaptive alarm system for visually impaired users using **STM32** & **embedded C**, integrating environmental sensing & multi-device alert signaling
- Designed & validated custom schematics & **PCB layouts** in **KiCad**, addressing power distribution, signal routing, & hardware verification through DRC/ERC checks.
- Implemented **UART**-based communication between two **STM32** microcontrollers, coordinating power control & alert signaling to support modular sensing & notification behavior.

FDM Infill Error Correction | *Python, PrusaSlicer, FullControl, Onshape* Mar.2024

- Designed Python algorithms to detect & correct 3D-print infill anomalies, utilizing **Prusa** FDM printers for algorithm validation.
- Processed binary scans with **OpenCV** thresholding & contour detection to locate defect regions.
- Reconstructed meshes via Delaunay triangulation & generated optimized G-code with **PrusaSlicer** & **FullControl**.